SREEANNAPURNA RALLABHANDI
AI/ML ENGINEER | GENERATIVE AI, LLM APPLICATIONS & AI AGENTS | ML DEPLOYMENT
Dallas, TX(Open to Relocation) | 940-377-2198 | sarallabhandi@gmail.com

PROFESSIONAL SUMMARY

AI/ML Engineer with 6+ years across Python engineering, applied machine learning, and AI applications. Build, train, evaluate, and deploy machine learning (ML) models, and develop production-oriented LLM applications and AI agents. Built a 0-to-1 agentic AI analytics solution with LangGraph, retrieval-augmented generation (RAG), FastAPI, and locally hosted Mistral 7B. Experienced in TensorFlow, Scikit-learn, inference APIs, model evaluation, Docker, AWS, and cross-functional delivery focused on AI reliability, accuracy, performance, and scalability.

TECHNICAL SKILLS

Machine Learning & Deep Learning: Python, Scikit-learn, TensorFlow, Keras, PyTorch, supervised and unsupervised ML, Random Forest, Isolation Forest, PCA, GRU, feature engineering, model training, validation and accuracy evaluation
Generative AI & LLM Applications: Generative AI, large language models (LLMs), AI agents, LangGraph, LangChain, RAG, Mistral 7B, retrieval, prompt engineering, tool calling, response grounding, LLM evaluation pipelines, reliability testing
Production ML & MLOps: ML model deployment, scalable model-serving and inference APIs, FastAPI, REST APIs, model versioning, experiment tracking, Docker, Kubernetes, AWS, GCP, CI/CD, Git, observability and performance monitoring
Data, NLP & Analytics: SQL, Pandas, NumPy, PySpark, time-series forecasting, anomaly detection, TF-IDF, sentiment classification, error analysis, data quality and validation

PROFESSIONAL EXPERIENCE

Invences Inc| Dallas, TX
AI Engineer - Agentic AI Analytics| Feb 2026 – Present

• Built a 0-to-1 enterprise AI analytics application with Python, Lang Graph, Fast API, retrieval-augmented generation (RAG), and locally hosted Mistral 7B to convert natural-language questions into data-grounded analytical responses.
• Developed REST inference and orchestration interfaces that coordinate agent state, data retrieval, tool selection, and response validation across operational, customer, and event data.
• Designed typed analytical tools and controlled execution paths that limit model-generated actions to approved data sources, calculations, and business rules.
• Implemented role-based access controls, query-level audit trails, and source-grounding checks so stakeholders can trace answers to retrieved evidence and authorized tool outputs.
• Developed LLM evaluation scenarios covering retrieval relevance, groundedness, tool-call correctness, response quality, and unexpected edge cases with QA and LLMOps teams.
• Added structured execution traces and error handling for agent decisions, tool inputs/outputs, and failed validations to accelerate fault isolation and reliability testing.
• Integrated Vector.dev, Victoria Metrics, and Grafana to observe edge/on-premises services, inspect telemetry and performance signals, and support troubleshooting for AI-enabled workflows.
• Collaborated with product-facing stakeholders, integration engineers, and QA to translate early-stage use cases into acceptance criteria, repeatable tests, API requirements, and release-ready workflows.

University of North Texas | Denton, TX
Graduate Research Assistant - Applied AI/ML| Sep 2023 - May 2025

• Built Python and PySpark preprocessing and feature-engineering pipelines for high-frequency UAV sensor telemetry, creating model-ready spatial, temporal, and environmental signals for training and evaluation.
• Trained and validated supervised and unsupervised machine learning models for anomaly and flight-deviation detection, achieving up to 96% validation accuracy in research experiments.
• Containerized ML workloads with Docker and deployed models on AWS for analytical inference, using Kubernetes and CI/CD practices to support repeatable training, testing, and releases.
• Performed error analysis comparing planned flight trajectories with observed telemetry, evaluating deviation patterns and refining features for more reliable detection of anomalous conditions.
• Developed Kafka, MongoDB, Flask, and REST API components connecting streaming telemetry ingestion, validated feature inputs, and downstream model experimentation.
• Implemented data-quality checks across sensor readings, route-intent events, and environmental inputs to reduce

malformed records entering experiments and improve repeatability of model evaluations.
• Supported an integrated mobility analytics pipeline handling multiple concurrent data streams and device feeds, applying real-time checks to investigate differences between intended and actual movement

FORSYS INC | Hyderabad, India

Data Analyst II | Mar 2022 - Aug 2023

• Built Python and SQL data services and ingestion pipelines for production analytics and downstream ML applications, introducing validation rules to improve feature and reporting quality.
• Translated business logic into automated reconciliation and exception workflows, reducing a manual reconciliation process from four hours to five minutes.
• Refined CRM data-quality controls and analytics logic with operational stakeholders, contributing to a 15% improvement in CRM accuracy and more consistent downstream datasets.
• Defined integration test cases, acceptance criteria, and exception-handling workflows with QA and business teams, contributing to a 60% reduction in data-quality tickets.
• Used Python, SQL, and exploratory analysis to uncover workflow bottlenecks and produce metrics that supported process optimization and data-informed application changes.
• Optimized SQL queries and reporting data models so analysts and engineers could work with CRM and operational data more efficiently.
• Examined sales-funnel trends with Python and SQL, found where leads were dropping off, and shared findings that contributed to an 8% increase in conversion.

VIDCENTUM SOFTWARE PVT. LTD | Hyderabad, India

Data Analyst | Jan 2020 - Dec 2021

• Developed Python, SQL, and distributed data pipelines for operational analytics, consolidating inputs from multiple IoT and application sources into governed reporting datasets.
• Improved data freshness through incremental ingestion and reusable transformations, reducing reporting latency and preparing consistent datasets for analytics and machine-learning consumption.
• Optimized PySpark processing for growing data volumes and monitored pipeline failures and data-quality exceptions to maintain dependable downstream reporting.
• Partnered with engineering and business teams to document metric definitions and produce reliable operational KPIs, enabling more consistent cross-functional decisions.
• Added SQL validation checks to flag missing or inconsistent data in incoming IoT feeds before it reached reporting tables.
• Published Snowflake and Redshift datasets supporting 75+ operational KPIs, giving teams consistent data to compare performance across product lines

EDUCATION

University of North Texas - Master of Science, Advanced Data Analytics (Artificial Intelligence Concentration) | Dec 2025
University of North Texas - Master of Science, Interdisciplinary Studies (Applied Analytics) | Dec 2024
ACE Engineering College - Bachelor of Engineering, Computer Science Engineering | May 2019